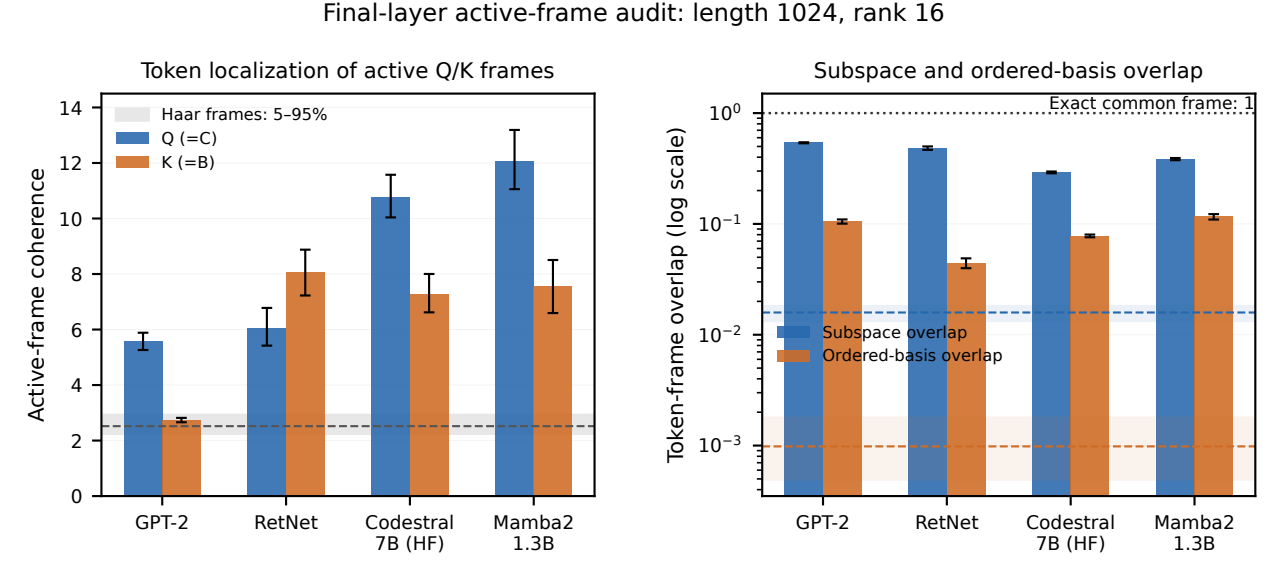
Final-layer active-frame audit: length 1024, rank 16
Token localization of active Q/K frames
Active-frame coherence
Haar frames: 5–95%
Q (=C)
K (=B)
0
2
4
6
8
10
12
14
GPT-2
RetNet
Codestral 7B (HF)
Mamba2 1.3B
Subspace and ordered-basis overlap
Token-frame overlap (log scale)
Exact common frame: 1
10^{0}
10^{-1}
10^{-2}
10^{-3}
Subspace overlap
Ordered-basis overlap
GPT-2
RetNet
Codestral 7B (HF)
Mamba2 1.3B
Bars ±95% document-bootstrap CI. Dashed lines/bands: Haar means/5–95% frame variation.